

Western Ghats Deforestation & Land Cover Change Detection

Region: **Nagarhole** | Years: 2024 → 2026

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Alert Level: Low Risk

Forest Cover Summary

Metric	Value
Forest Cover (Year 1)	31.45%
Forest Cover (Year 2)	34.20%
Forest Loss	0.00%
Forest Gain	2.75%
Net Forest Change	+2.75%

Land Cover Percentages

Class	Year 2024 %	Year 2026 %	Change %
Forest	31.45%	34.20%	+2.75%
Agriculture	0.02%	0.01%	-0.01%
Urban	31.67%	39.43%	+7.75%
Water	34.42%	20.27%	-14.15%
Barren	2.44%	6.10%	+3.66%

Key Land Cover Transitions

Transition	Area (%)
Forest → Urban	2.701%
Forest → Agriculture	0.002%
Forest → Barren	0.079%
Agriculture → Urban	0.000%

Recommendations

- Urban expansion detected near forest boundary.

Academic Analysis & Ecological Justification

1. Reasoning for Land Cover Shifts

A positive trend with a 2.75% increase in forest cover suggests effective local conservation efforts and natural regeneration, aligning with recent afforestation drives under the Green India Mission. The 7.75% expansion in built-up urban areas is a direct consequence of rapid demographic shifts and infrastructure development. Recent environmental appraisals emphasize that unregulated urbanization in the eco-sensitive zones of the Western Ghats severely fragments primary habitats. A concerning reduction of 14.15% in water bodies correlates with reduced groundwater recharge due to catchment degradation, exacerbating hydrological vulnerability in the region as noted by the Central Water Commission. The 3.66% increase in barren land highlights soil erosion and localized resource extraction (such as quarrying), which strip the land of its natural ecological resilience.

2. Steps for Ecological Balance

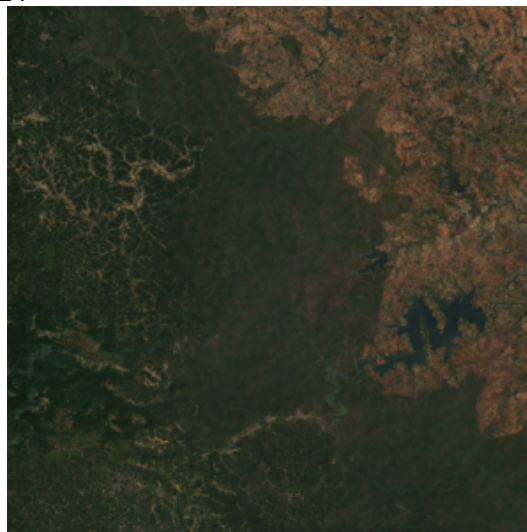
- **Enforcement of Eco-Sensitive Zones (ESZs):** Strict implementation of zoning laws as recommended by the Kasturirangan Committee to restrict hazardous industrial and mining activities in vulnerable tracts.
- **Agroforestry Integration:** Transitioning from monoculture plantations to biodiversity-friendly agroforestry systems in buffer areas to restore soil fertility and create wildlife corridors.
- **Hydrological Restoration:** Rejuvenating degraded catchments through watershed management and afforestation to secure the water table and sustain regional hydrology.

3. Conclusive Points

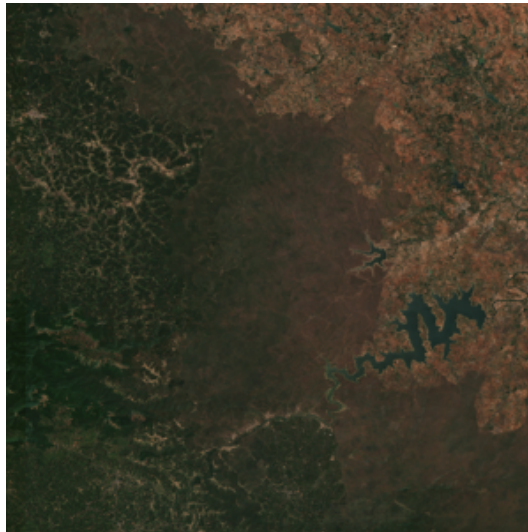
- **Habitat Fragmentation & Edge Effects:** The observed transitions in Nagarhole reveal a concerning pattern. The conversion of natural habitats to anthropogenic land uses isolates endemic species and diminishes overall biodiversity capacity.
- **Ecological Vulnerability:** The land-cover dynamics underscore a precarious balance. Without proactive management, even minor anthropogenic perturbations could trigger irreversible ecological decline in this global biodiversity hotspot.
- **Imperative for Data-Driven Policy:** Real-time, satellite-derived insights emphasize the critical need for executing ground-level environmental regulations promptly. Continued unmonitored land-cover alterations will irreversibly compromise the long-term ecological and economic stability of peninsular India.

Visual Maps

Satellite Image — Year 2024



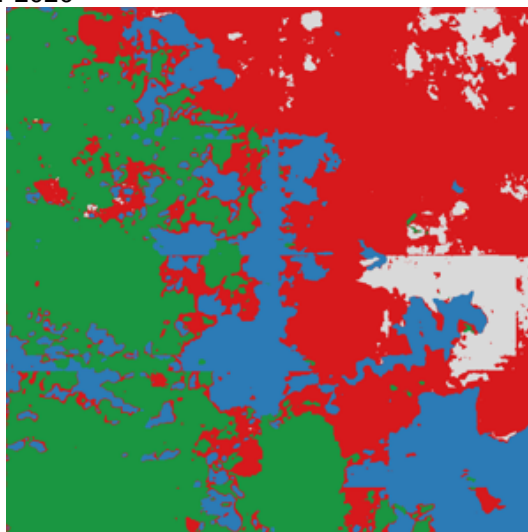
Satellite Image — Year 2026



Segmentation Map — Year 2024



Segmentation Map — Year 2026



Change Detection Map (Red=Forest Loss, Green=Forest Gain, Orange=Urbanisation)

